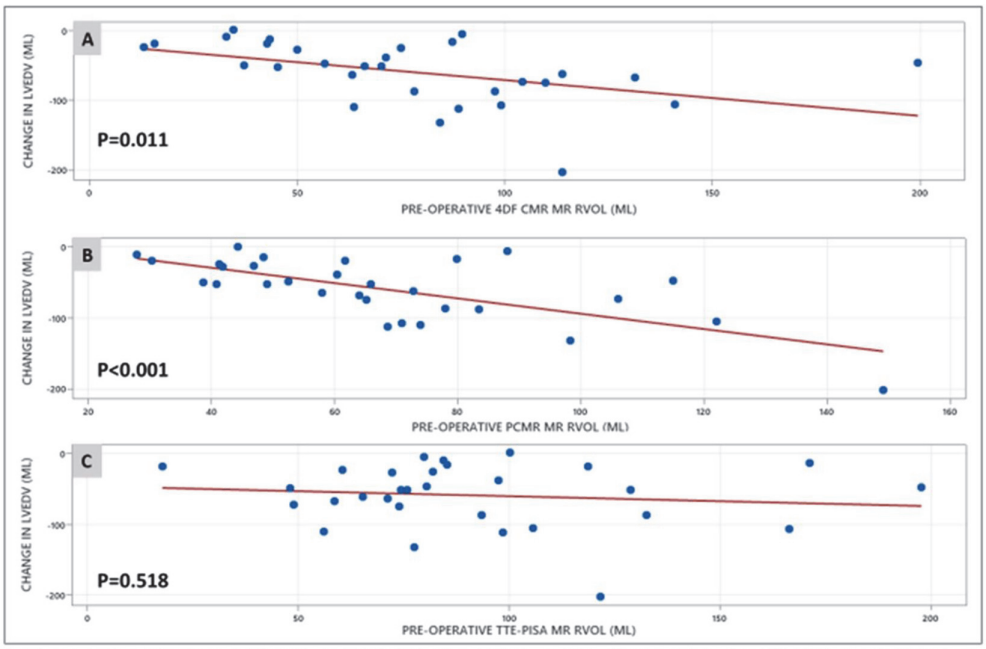

Linear regression analysis between post-operative change in left ventricular end-diastolic volume and pre-operative mitral regurgitant volume. Panel (A) presents MRegurg volume quantified by 4DF CMR, panel (B) by PCMR and panel (C) by the TTE-PISA method.

4DF CMR=4D flow cardiovascular magnetic resonance imaging; LVEDV=left ventricular end-diastolic volume; MRegurg=mitral regurgitation; PCMR=phase-contrast magnetic resonance imaging; RVOL=regurgitant volume; TTE-PISA=transthoracic echocardiography-proximal isovelocity surface area method.

Abstract 169 Figure 2 Linear regression analysis between post-operative change in left ventricular end-diastolic volume and pre-operative mitral regurgitant volume
Panel (A) presents MR regurgitant volume quantified by 4DF CMR, panel (B) by PCMR and panel (C) by the TTE-PISA method
4DF CMR=4D flow cardiovascular magnetic resonance imaging; LVEDV=left ventricular end-diastolic volume; MR=mitral regurgitation; PCMR=phase-contrast magnetic resonance imaging; RVOL=regurgitant volume; TTE-PISA=transthoracic echocardiography-proximal isovelocity surface area method

regurgitant volume was quantified by PCMR ($p<0.001$) and 4DF CMR ($p=0.011$), but not when assessed by TTE ($p=0.518$) (Figure 2).

Conclusion This study demonstrates the potential clinical utility of 4DF CMR in the assessment of degenerative mitral regurgitation.

Conflict of Interest None

170 NON-INVASIVE ASSESSMENT OF STROKE WORK USING CARDIAC MAGNETIC RESONANCE IMAGING

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Introduction Although widely adopted in clinical practice, the left ventricular ejection fraction (LVEF) has significant limitations in heart failure assessment1. Gold standard physiological measures of heart function utilise pressure-volume (PV) loops, particularly the stroke work (SW), represented as the area bounded by the loop. Invasive assessment of PV-loops precludes its wide-spread clinical use. Cardiac magnetic resonance (CMR) imaging allows the accurate assessment of volumetrics and recent studies have found it useful in estimating left ventricular filling pressure2. We hypothesise that non-invasive parameters can be used to accurately estimate SW.

Methods In the Wellcome Trust PREFER-CMR study (NCT05114785) invasive PV-loops were collected using impedance catheters (7-Fr PV-Loop Catheter CA-71103-PL; CD Leycom, NL) from 8 patients who also underwent CMR assessment within 4 hours, which were then analysed using custom algorithms in MATLAB. A non-invasive, physiological method of SW assessment was developed using CMR parameters and the peripheral blood pressure. This method was subsequently applied to a large retrospective cohort of 835 heart failure and pulmonary hypertension patients who had undergone both CMR and invasive pressure assessment at the Sheffield Pulmonary Vascular Disease Unit2. Correlations were assessed using the Pearson Correlation Coefficient. Variables were normalised prior to backward conditional multivariate Cox regression. All work was received ethical approval (REC: 21/NE/0149, 16/YH/0352).

Results CMR SW correlated with that from invasive PV-loops (typical loop shown in Figure 1) in the small discovery impedance catheter cohort ($r=0.861$, $P=0.006$) and no significant difference demonstrable (Mean invasive SW 1.298±0.13J vs. CMR SW 1.228±0.14J, paired t-test $p=0.350$). Non-invasive CMR SW significantly correlated with hybrid invasive/CMR-derived PV-loop SW ($r=0.995$, $p<0.001$). Univariate regression analysis demonstrated significant risk of all cause death associated with reduced CMR SW, increased CMR-derived pulmonary capillary wedge pressure (PCWP) and reduced LVEF, with CMR SW having the greatest effect size (Table 1). On multivariate regression, age, male sex, CMR SW and CMR PCWP remained independently predictive of overall death whilst LVEF did not (Table 1). For 1-year mortality, only CMR SW age and male sex remained significant, with CMR-derived PCWP and LVEF excluded (Table 2). ROC curve analysis gave an area under the curve of 0.634 for

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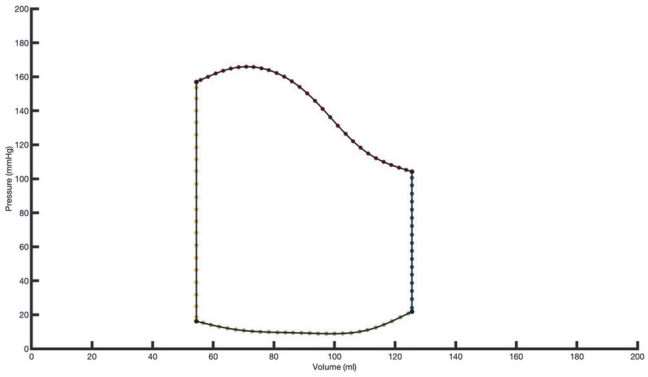
Abstract 170 Table 1 Results of both univariate and multivariate Cox regression analysis for overall mortality. *Denotes significant at p<0.05

	Overall Mortality			
	Univariate Regression		*Multivariate Regression*	
	HR (CI 95%)	*p-value*	*HR (CI 95%)*	*p-value*
Age	*1.049 (1.039-1.059)*	<0.001*	21.691 (10.457-44.993)	<0.001*
Sex (male)	*1.474 (1.212-1.794)*	<0.001*	1.545 (1.253-1.906)	<0.001*
BSA (m^2)	*0.630 (0.422-0.941)*	0.024*	0.569 (0.251-1.286)	0.175
CMR PCWP	1.046 (1.020-1.074)	<0.001*	3.368 (1.615-7.021)	0.001*
CMR SW	0.352 (0.261-0.474)	<0.001*	0.045 (0.019-0.107)	<0.001*
LVEF (%)	0.988 (0.979-0.996)	0.005*	2.066 (0.917-4.654)	0.08

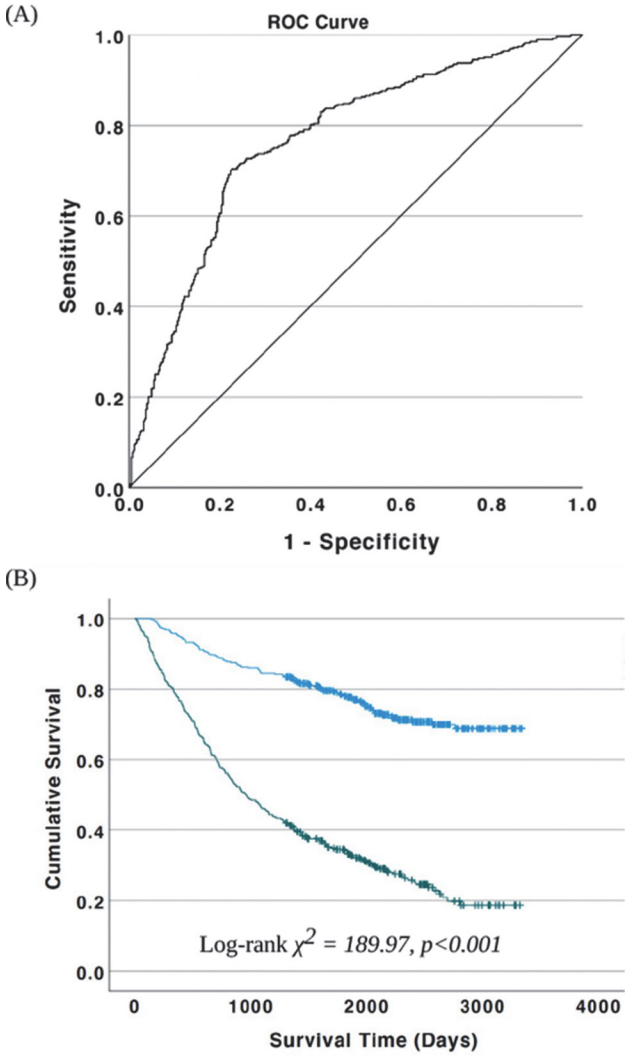
Abstract 170 Table 2 Results univariate and multivariate Cox regression analysis for overall mortality. *Denotes significant at p<0.05

	1-Year Mortality			
	Univariate Regression		*Multivariate Regression*	
	HR (CI 95%)	*p-value*	*HR (CI 95%)*	*p-value*
Age	1.040 (1.021-1.060)	<0.001*	12.340 (3.241-46.979)	<0.001*
Sex (male)	1.798 (1.232-2.624)	<0.001*	2.210 (1.412-3.458)	<0.001*
BSA (m^2)	0.463 (0.213-1.003)	0.051	0.237 (0.050-1.116)	0.069
CMR PCWP	1.005 (0.952-1.061)	0.866	2.587 (0.568-11.778)	0.219
CMR SW	0.230 (0.127-0.419)	<0.001*	0.048 (0.011-0.209)	<0.001*
LVEF (%)	0.990 (0.974-1.006)	0.220	2.649 (0.580-12.107)	0.209



Abstract 170 Figure 1 A typical invasive Pressure Volume Loop obtained from the impedance catheter. Successive beats have been averaged and subsequent data fitted using spline functions. Volumes were normalised to CMR derived end-diastolic and end-systolic volumes

CMR SW in isolation and 0.767 for the multivariate model (Figure 2A) for overall mortality. Using the multivariate regression, a risk score based on age, sex, CMR SW and CMR PCWP with cut-off value of >1.45 is shown in Figure 2B demonstrating significant difference in survival.



Abstract 170 Figure 2 (A) ROC curve analysis for the multivariate model produced for overall mortality. (B) Kaplan-Meier curve for multivariate risk score with cut off (determined from ROC curve) of >1.45. The blue curve shows a low-risk score and the green a high-risk score. Censored events are shown by vertical dashes

Conclusions A combination of simple blood pressure assessment and CMR-derived volumetrics were able to accurately estimate the cardiac SW which was predictive of both 1-year and overall mortality. Such a method is amenable to high throughput and artificial intelligence assisted clinical application for heart failure management.

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Conflict of Interest Nil

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